Lab_4_VibeCoding...
PDF

I've received your **Lab 4: VibeCoding – Student Handout** PDF.

I'll wait for your instructions.

https://github.com/SETAPESU26/29_digDug

wait for my instructions

Got it. I have the 29_digDug GitHub repository ↗ and your Lab 4 VibeCoding handout. I'll wait for your instructions before proceeding.

what the bugs are and help me install pygame

Worked for 14s

I've checked the actual `game.py` and `README.md` in your 29_digDug repository ↗. Let's first get Pygame installed, then I'll explain the bugs you need to fix for Lab 4.

## 1. What are the bugs in Dig Dug?

The README specifies **one deliberate bug and three optional features**. pygame wiki

### Task 1 — Fix the enemy pathfinding bug

In `game.py`, find this function:

```python
def bfs_path(grid, start, goal):
    queue, parents = deque([start]), {start: None}
    while queue:
        cell = queue.pop()  # BUG IS HERE
```

### What's wrong?

The game uses breadth-first search (BFS) to find the shortest path through cleared tunnels. But `queue.pop()` removes the newest item, making the search behave like depth-first search instead.

The fix is to change that line to:

```python
```

```
cell = queue.popleft()
```

This makes the search process the oldest queued cell first, allowing BFS to find the shortest path in the unweighted grid.

### Tasks 2–4 — Implement the optional features

These three functions currently contain `pass`, meaning their behavior hasn't been implemented.

| Function | What you need to add |
|---|---|
| `dirt_color(row)` | Different colors for different underground depth bands. |
| `on_enemy_popped(enemy, score)` | An effect when an enemy bursts, such as bonus points or a visual effect. |
| `enemy_speed_multiplier(level)` | Make enemies move faster as the level increases. |

These are separate tasks, so you should make a separate Git commit for each task, as required by your Lab 4 handout.

## 2. Install Pygame on your Windows laptop

Your earlier attempts failed because you're using **Python 3.14**, and the Pygame release available on PyPI lists Windows wheels for Python 3.13 but not Python 3.14 in the results I checked. Installing Python 3.13 alongside 3.14 is the simplest approach. PyPI +1

### Step 1 — Install Python 3.13



**Download Python 3.13 for Windows**

Open the official Python for Windows downloads page ↗.

Find Python 3.13 and download the **Windows installer (64-bit)** if your Windows installation is 64-bit.

You can keep Python 3.14 installed. You don't need to uninstall it.

### Step 2 — Open your project in VS Code

Open the terminal using **Terminal → New Terminal** and enter these commands one at a time.

Your existing project folder is:

‹/› powershell

```
cd "C:\Users\sudha\OneDrive\Documents\29_digDug"
```

Check that Python 3.13 is available:

```powershell
py -3.13 --version
```

You should see something similar to:

```
Python 3.13.x
```

If it says that no suitable Python runtime was found, finish installing Python 3.13 first.

### Step 3 — Install Pygame

Run:

```powershell
py -3.13 -m pip install --upgrade pip
```

Then:

```powershell
py -3.13 -m pip install pygame
```

Wait for the installation to finish.

### Step 4 — Verify that Pygame works

Run:

```powershell
py -3.13 -c "import pygame; print(pygame.version.ver)"
```

If it prints a Pygame version without an error, the installation is successful.

**Step 5 — Run your Dig Dug game**

Run:

```powershell
py -3.13 game.py
```

You should see the game window. Before modifying anything, record the required **10-second video of the original game**, as your Lab 4 handout instructs.

## 3. Let's do this step by step

First, complete Step 1 and then run this command in your VS Code terminal:

```powershell
py -3.13 --version
```

**Send me a screenshot of the output.** I'll guide you through the next step based on what appears.

yes i am done

before.mp4
File

Each task must be completed using an iterative process involving LLM suggestions and your critical code review.

### Task 1: Fix the enemy pathfinding bug

Once a tunnel connects an enemy to the player, the enemy is supposed to take the *shortest* route through cleared cells, using breadth-first search. In the current build, enemies do eventually reach the player, but along long, winding routes instead of the shortest one — the search behaves like depth-

first search rather than breadth-first. Look at which end of the `deque` the search removes cells from in `bfs_path`.

done

### Task 2: Implement `dirt_color(row)`

Called once per dirt cell per frame in `draw`, as `shade = dirt_color(row) or (170 - row * 6, 110 - row * 4, 60)`. It receives the cell's row index and should return an `(r, g, b)` color, or `None` to keep the default gradient (which darkens with depth). Idea: give each row band a distinct color instead of a smooth gradient.

### Task 3: Implement `on_enemy_popped(enemy, score)`

Called from `pump()` right after an enemy reaches its fourth inflation stage, is removed from play, and its points are added. It receives the `Enemy` that popped and the score after its points were added. Its return value is ignored. Idea: a particle burst at `enemy.cell`, or extra points for popping an enemy deep underground.

### Task 4: Implement `enemy_speed_multiplier(level)`

Called every time a non-locked enemy finishes a step, as part of `enemy.step_timer = ENEMY_DELAY * (1.6 if enemy.ghost else 1) / (enemy_speed_multiplier(self.level) or 1)`. It receives the current level number and should return a speed multiplier, or `None` for the default speed (1x). A higher multiplier makes enemies move more often (shorter delay between steps). Idea: return `1 + 0.1 * level` so enemies get gradually faster on deeper levels.



**after.mp4**
File

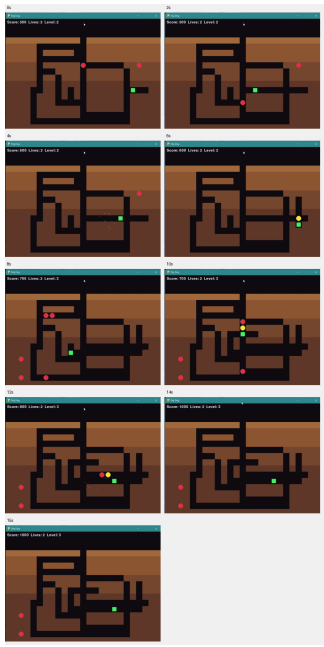
confirm if this correct

Worked for 14s

Analyzed

Analyzed

**Analysis output 1**



‹/›

yes i have verified as i reach a higher level the speed increases and enemies change direction and find the shortest path when i am near them

are all my deliverables complete?

Worked for 7s

yes i have pushed it in my githib and have dont 4 git commits seperately and is under uploaded under lab 4
chat history is yet to be done

is this chat history required for this lab?